Homework 1

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Chapter 0

Problem 5

If C is a set with c elements, how many elements are in the power set of C? Explain your answer.

There are $2^{|c|}$ elements in the power set of C. This is because the power set C is the set of every possible subset of C.

Problem 7

For each part, give a relation that satisfies the condition.

a. Reflexive and symmetric but not transitive
$S = \{1, 2, 3\}$
$R = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (3, 1), (3, 2), (3, 3)\}$
b. Reflexive and transitive but not symmetric
$S = \{1, 2, 3\}$
$R = \{(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$
c. Symmetric and transitive but not reflexive
$S = \{1, 2, 3\}$
$R = \{(1, 2), (1, 3), (2, 1), (2, 3), (3, 1), (3, 2)\}$

Problem 10

Find the error in the following proof that $2 = 1$. Consider the equation $a = b$. Multiply both sides by a to obtain $a^2 = ab$. Subtract b^2 from both sides to get $a^2 - b^2 = ab - b^2$. Now factor each side, $(a + b)(a - b) = b(a - b)$, and divide each side by $(a - b)$ to get $a + b = b$. Finally, let a and b equal 1, which shows that $2 = 1$.

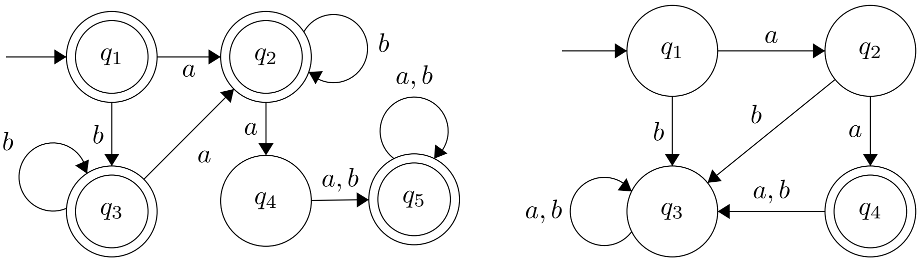
The error in this proof lies in the fact that letting $a = 1, b = 1$ needs to hold true for every part of the proof. For example, $a = 1, b = 1$ holds true for the first 3 steps of the proof. However, when stating that one should divide to obtain $a + b = b$, you would actually by dividing by zero, which is undefined. Thus contradicting the original proof.

Chapter 1

Problem 1.5g

Each of the following languages is the complement of a simpler language. In each part, construct a DFA for the simpler language, then use it to give the state diagram of a DFA for the language given. In all parts, $\sum = \{a, b\}$.

Figure 1: $\{w|w$ is any string that doesn't contain exactly two $a's\}$



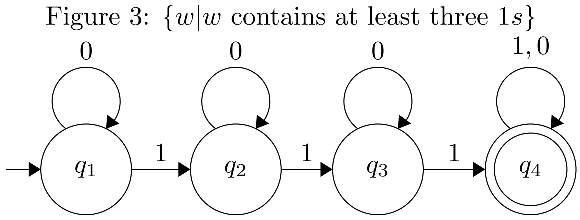
(a) Exactly 2 a's

(b) Only 2 a's

Problem 1.6b

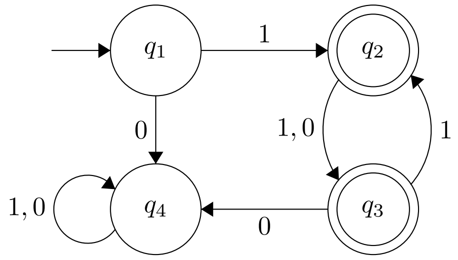
Give state diagrams of DFAs recognizing the following languages. In all parts, the alphabet is $\{0, 1\}$.

Figure 3: $\{w|w$ contains at least three $1s\}$



Problem 1.6i

Figure 4: $\{w|$ every odd position of w is a $1\}$



Problem 1.14a

Show that if M is a DFA that recognizes language B, swapping the accept and nonaccept states in M yields a new DFA recognizing the complement of B. Conclude that the class of regular languages is closed under complement.
Suppose we have DFA M_1 which recognizes Language B, where $B = \{w | w \text{ is an odd number of 1s }\}$

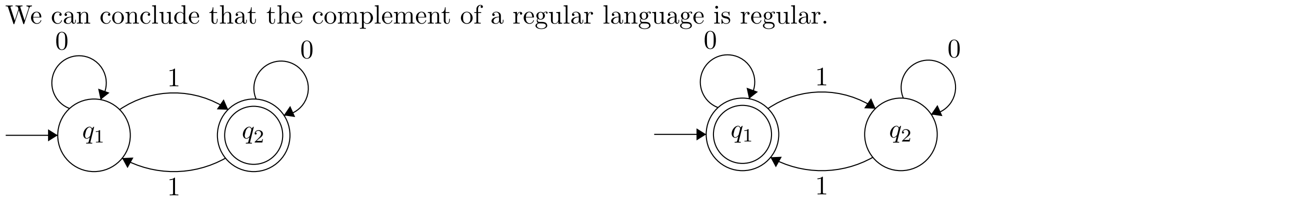
The Formal definition of B would be:

- $Q = \{q_1, q_2\}$
- $\sum = \{0, 1\}$
- $\delta(q_1, 0) = q_1, \delta(q_1, 1) = q_2, \delta(q_2, 0) = q_2, \delta(q_2, 1) = q_1,$
- $q_0 = q_1$
- $F = q_2$

Taking the Complement of B we get the following definition for a new machine M_2

- $Q = \{q_1, q_2\}$
- $\sum = \{0, 1\}$
- $\delta(q_1, 0) = q_1, \delta(q_1, 1) = q_2, \delta(q_2, 0) = q_2, \delta(q_2, 1) = q_1,$
- $q_0 = q_1$
- $F = q_1$

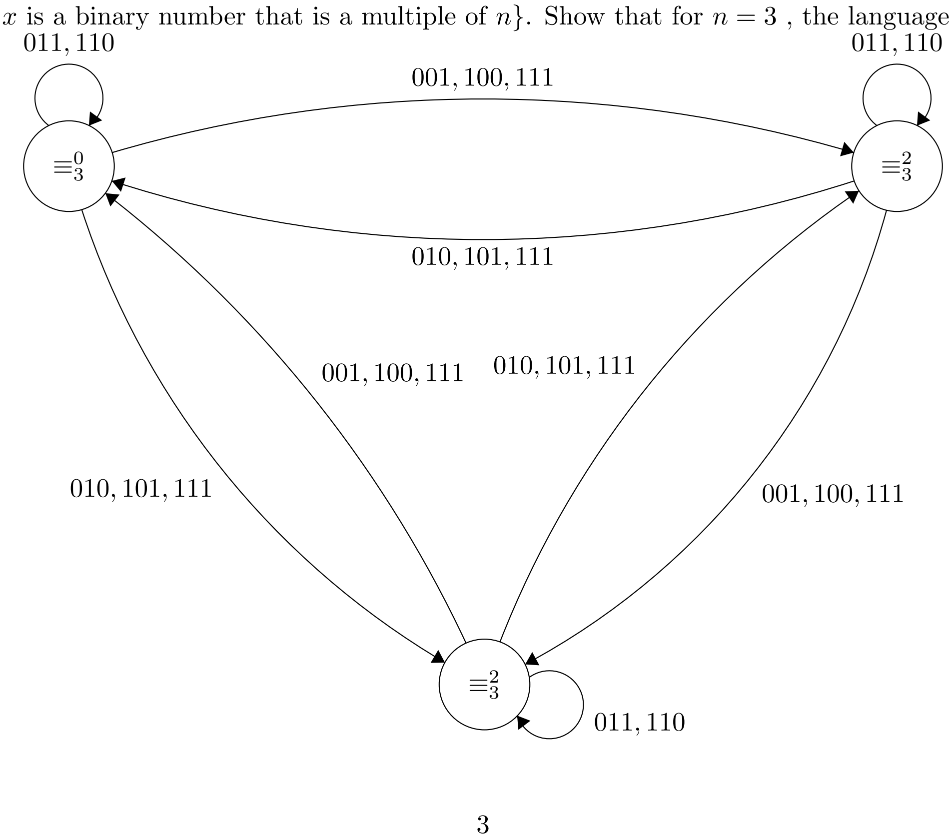
We can conclude that the complement of a regular language is regular.



(a) B (b) $\overline{B}$

Problem 1.37

Let $C_n = \{x | x \text{ is a binary number that is a multiple of } n\}$. Show that for $n = 3$, the language C_n is regular.



Problem 1.48

Let $\sum = \{0, 1\}$ and let
$D = \{w|w$ contains an equal number of occurrences of the substrings 01 and 10$\}$.

Thus $101 \in D$ because 101 contains a single 01 and a single 10, but $1010 \notin D$ because 1010 contains two 10s and one 01. Show that D is a regular language.